

Average (conclusion)

① ⇒ Formula based

average of natural no.

(Find the average of First 10)

$$\text{First 7} = \frac{7+1}{2} = \frac{8}{2} = 4$$

$$\frac{(n+1)}{2}$$

1st Type

Formula based

average of even no.

$(n+1)$ total terms

average of odd no.

↓ n

AP series
(equal diff.)

First four even no.
(5)

first 10 odd no.
(10)

First term + last term

Note - odd series
↳ middle value (average)

even series
↳ middle 2 values (sum)

$$\begin{aligned} & \text{No. of terms} \\ & = \frac{36 + 12}{2} \\ & = \frac{24}{2} \\ & = 12 \end{aligned}$$

or
First + last
2

(if series is consecutive or AP series)

middle term is always the average

→ average = 52

48 50 (52) 54 56

2nd Type

Inclusion

Exclusion

i) average of group

↓

increase / decrease

↓

When new person ~~is~~ is added ~~on~~ left the group

added

$$\begin{array}{ccccc} 25 & 25 & 25 & 25 & 25 \\ \hline -2 \times 5 & & & & \end{array}$$

$$\Rightarrow -10$$

$$\begin{array}{r} 27 \\ \hline 37 \end{array} \quad \begin{array}{l} \text{new person} \\ \text{average} \\ +2 \end{array}$$

age

ii) average of group

↓

rise / ↓ se

↓

When ~~new~~ person left the group

removed

$$\begin{array}{cccc} 25 & 25 & 25 & 25 \\ \hline +2 \times 4 & & & \end{array} \quad \boxed{25} \quad \text{X (left)}$$

↓ se

average
↓
+2

average
↓
-2

$$+8 \Rightarrow 25 - 8 = 17$$

$$\hookrightarrow 4 \times 2 = -8$$

$$\hookrightarrow 25 + 8 \Rightarrow 33$$

000
111)

Sometimes they don't give value of new team, they give you how much % ↑se or ↓se. Then ask old or new average?

group average
↑se by 2

$$\begin{array}{ccccc} x & x & x & x & x \end{array} \quad \boxed{30} \rightarrow \text{new person age (value)}$$

$$\begin{array}{ccccc} x+2 & x+2 & x+2 & x+2 & x+2 \end{array} \quad \boxed{30}$$

$$5x + 10 = +10 \checkmark$$

$$30 - 10 = 20$$

$$\downarrow$$

new average

$$\text{old} = 20 - 2 = 18$$

iv) given average of certain people in group is 25

a new person is added to the group and group average having the age 20 is ~~25~~. Thereby, group average is ~~increased~~ by ↓se by 1. Find the total no. of person in a group.

$$\frac{25}{24} \quad \frac{25}{24} \text{) } (1 \quad \Rightarrow \quad 25 - 20 = 5$$

group 25 (avg)

$$\left(\frac{5}{1} \right) \rightarrow 5 \checkmark$$

3rd type

Combined average

{ average of 1st group
average of 2nd group
Find combined average? }

Group 1 (average)

Group 2 (average)

Average / No. of Person
in Group

Average / No. of Person
in a group

Class $\rightarrow$ 60 students $\begin{cases} 40\% \text{ boys} \\ 60\% \text{ girls} \end{cases}$

B
boys

G
girls

12 / 20
No. avg.

16 / 30
No. avg.

Find combined
average of
class

(I)
Assume
Method

-5×12
 $= -60$

$+5 \times 16 = +80$

Total students
 $= 12 + 16 = 28$

25 $\rightarrow$ assumed i.e.
28

$+80 - 60$
 $= +20$ extra value

distribute
among 28 students

$+ \frac{20}{28} = +0.7$

add 0.7
to
assumed
one

$25 + 0.7$
 $= 25.7$

+ add
- sub

20 + 20

10

(II.)

given
exact
average

Average of class of 60 students is 30

Boys (avg) →

40%

Boys

avg → 20

Figure out average
of girls.

Total = 60 (avg = 30)

(40%) Boys = 24

girls = 36

B

G

$$\begin{array}{r} 24 \overline{) 240} \\ -10 \Rightarrow 10 \times 24 \\ = 240 \end{array}$$

$$36 \overline{) 360}$$

$$60 \overline{) 300}$$

$$24 \overline{) 240}$$

$$36 \overline{) 360}$$

$$\begin{array}{r} 240 \div 36 = 6.6 \end{array}$$

$$60 \overline{) 300}$$

combined

$$30 + 6.6 \Rightarrow 36.6$$

last type

average of 5 numbers = 30

10 20 30 40 50

avg of first two = 15
" " last two = 45

Find value of third one

Ans.

$$\begin{array}{ccccccc} 30 & 30 & 30 & 30 & 30 & & \\ \hline & & & & & +15 & \\ & & & & & 45 & \\ -15 \times 2 & & & & & & \\ -30 & & & & & & \\ & & 0 & & + 2 \times 15 & & \\ & & & & \Rightarrow 30 & & \end{array}$$

+30 -30

if +10
↓
Sub from
avg

if -10
↓
add to
avg